Product Data & Analytics Portfolio

Selected work samples for Product Data Analyst applications

TWO PROJECTS. ONE APPROACH.

I build practical prototypes that turn complex financial information into structured inputs, transparent calculations, and actionable outputs. These projects reflect independent exploration of data quality, workflow design, and user-facing analysis.

01 BUSINESSFINANCE

Small-business bookkeeping and financial-close MVP: transaction classification, rule-based journal entries, invoice/bill tracking, straight-line depreciation, financial statements, and exception review.

02 MARKETLENS

Options analysis prototype: scenario-based profit and loss, Greeks sensitivity, bid-ask spread checks, risk planning, and optional historical indicator analysis.

RELEVANT CAPABILITIES

Data validation / Structured classification / Calculation testing / Financial analysis / User-oriented reporting / Workflow documentation

Scope note: Both are independent prototypes, not production-deployed client systems.

BusinessFinance

AI-Assisted Financial Close MVP | Browser-based prototype

PROBLEM & APPROACH

Small businesses need to understand cash, revenue, outstanding invoices, expenses, and potential errors without navigating a general ledger. This prototype provides plain-language transaction inputs and uses explicit accounting rules to produce structured records.

IMPLEMENTED FEATURES

Transaction mapping

12 transaction types mapped to debit/credit accounts with basic checks.

Receivables & payables

Invoice and vendor-bill register with related payment matching and outstanding balances.

Asset schedules

Fixed-asset input and monthly straight-line depreciation calculations.

Reporting

Financial statement views, business KPIs, and current-month vs. prior-three-month comparisons.

Quality review

Exception flags and review guidance for missing categories or payment references.

DATA WORKFLOW

INPUT	PROCESS	OUTPUT
Transaction type, amount, date	Rule-based entries & matching	Reports, KPIs, exception flags

Limitations: Prototype does not fully automate tax, payroll liabilities, inventory costing, or bank reconciliation.

MarketLens

Options Analysis & Decision-Support Prototype

PROBLEM & APPROACH

Options traders must assess possible outcomes, sensitivity to market changes, liquidity, and downside risk. MarketLens organizes these considerations into a single interactive analysis workflow rather than relying on a single price forecast.

IMPLEMENTED ANALYSIS

Scenario modeling

Call/put profit-and-loss estimates across user-defined price and time scenarios.

Sensitivity

Greeks-based analysis to explore how key assumptions affect estimated outcomes.

Market quality

Bid-ask spread and liquidity checks to highlight execution considerations.

Risk planning

User-defined targets and risk limits to compare potential gains and losses.

Historical context

Optional analysis of historical market indicators; not a live market-data service.

RELEVANCE TO PRODUCT DATA ANALYSIS

Defines consistent inputs, applies repeatable calculations, compares scenarios, and presents outputs in a format designed for review. The emphasis is on transparent assumptions and testable logic.

Scope note: Research/education prototype; simulated outputs are not investment advice or verified live quotes.